B.TECH ARTIFICIAL INTELLIGENCE & MACHINE LEARNING · CA2 CAPSTONE PROJECT

Decision Debt OS

Turning business decisions into measurable learning



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Division: [Division] **Institution:** [Institution Name] **Academic Year:** 2026–27

THE PROBLEM

Decisions get made. Assumptions get forgotten.



Dashboards and BI tools show what happened. Project trackers show what is scheduled. Neither one records **what a team believed would happen** before they acted — so when the belief turns out to be wrong, nothing captures that it was ever tested, and the same unexamined assumption drives the next decision.

CORE INSIGHT

A decision is not finished when it is made.

> *“A business decision is incomplete until its original assumption is compared with the real-world outcome.”*

This gap has a name:

Decision Debt — the accumulation of decisions whose assumptions were never checked against reality.

DEFINING THE PROBLEM

What Is Decision Debt?

Unresolved Assumptions

A decision was made, but its underlying belief was never written down in a checkable form.

Overdue Reviews

A deadline for checking the outcome has passed, and no one has looked at the result.

Invalidated Assumptions Still Driving Work

The assumption was checked, found false, but the decision or policy it justified is still active.

Repeated Forecasting Errors

The same type of assumption keeps failing across different decisions, and no one notices the pattern.

THE SOLUTION

Capture → Structure → Forecast → Measure → Evaluate → Learn



UNSTRUCTURED INPUT

"Let's offer the 10% discount to high-risk customers — I think it'll cut churn to around 18% within a month."

STRUCTURED DECISION RECORD

INTERVENTION	10% retention discount
TARGET METRIC	30-day churn, high-risk cohort
FORECAST	25% → 18%
CONFIDENCE	62%
DEADLINE	+30 days

WORKED EXAMPLE

One Decision, Fully Specified

> *"We believe a 10% retention discount will reduce 30-day churn among high-risk customers from 25% to 18%."*

OWNER

Retention Lead, Growth Team

CONFIDENCE (FORECAST)

62% self-reported probability

INTERVENTION

10% discount, auto-applied at checkout

TARGET METRIC

30-day churn rate, high-risk cohort

DEADLINE

30 days from decision date

ESTIMATED COST

~$4.80 / customer (10% of avg. order)

KILL CONDITION

If interim churn > 25% at day 15, stop offer

ACTUAL RESULT

Pending — 30-day window not yet elapsed

Baseline shown here (25%) is one risk-tier definition; the dataset's observed high-risk cohort baseline is 37% — see Slide 7.

DATASET & PIPELINE — SYNTHETIC, FOR DEMONSTRATION ONLY

From Customer Record to Decision Evaluation

Customer profile → Churn-risk prediction → Retention decision → Treatment / control group → 30-day outcome → Decision evaluation

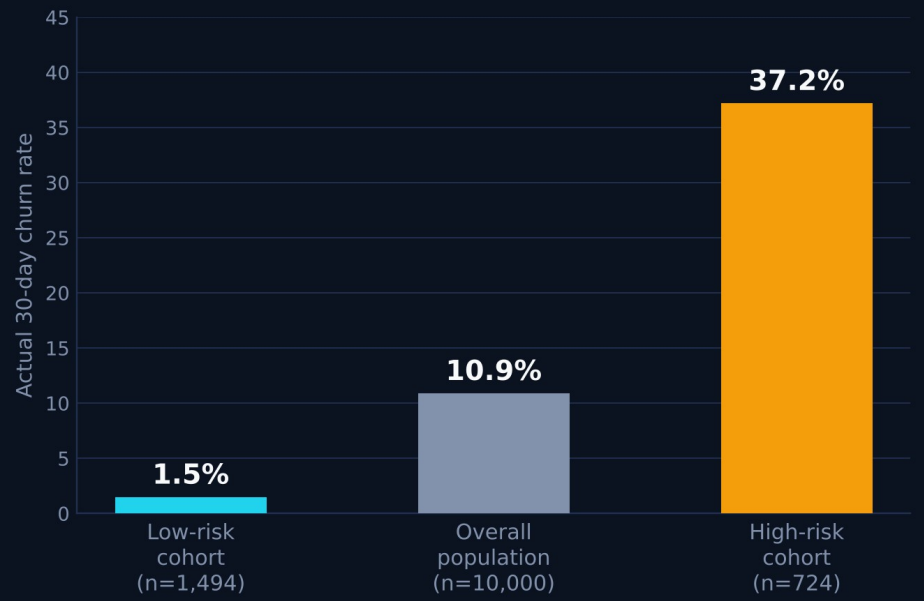
SYNTHETIC CUSTOMER-RETENTION DATASET

RECORDS 10,000 fictional customer rows

FIELDS age, gender, country, income, spending score, purchases, avg. order value, membership years, site visits/mo, cart-abandon rate, churn flag, feedback text, last purchase date

STATUS Clearly labelled synthetic data — used for demonstration only; not a real company's data

Real, observed churn rate by risk segment



MACHINE LEARNING COMPONENT

Predicting Risk, Not Proving Causation

HOW IT'S BUILT

TASK Binary classification — churned vs. retained

FEATURES Spending score, cart-abandon rate, site visits/mo, purchases, avg. order value, membership years, income, age, gender, country

PREPROCESSING Standardization (numeric) + one-hot encoding (categorical)

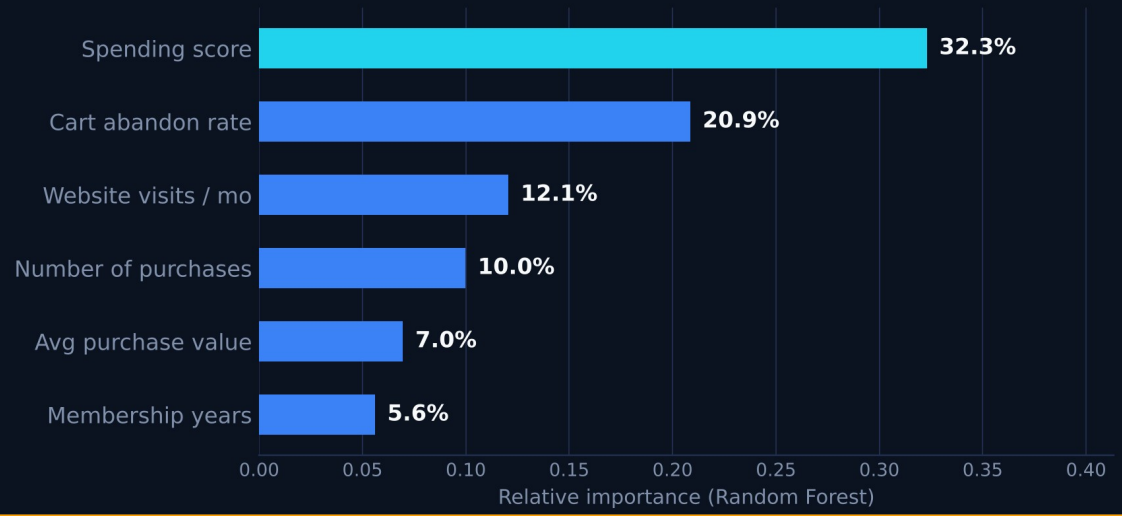
SPLIT 80 / 20 train-test, stratified by class

MODELS COMPARED Logistic Regression vs. Random Forest (300 trees)

0.26	0.78	0.39	0.84
Precision	Recall	F1-score	ROC-AUC

Random Forest · held-out test set · n = 2,000 (real results, not placeholders)

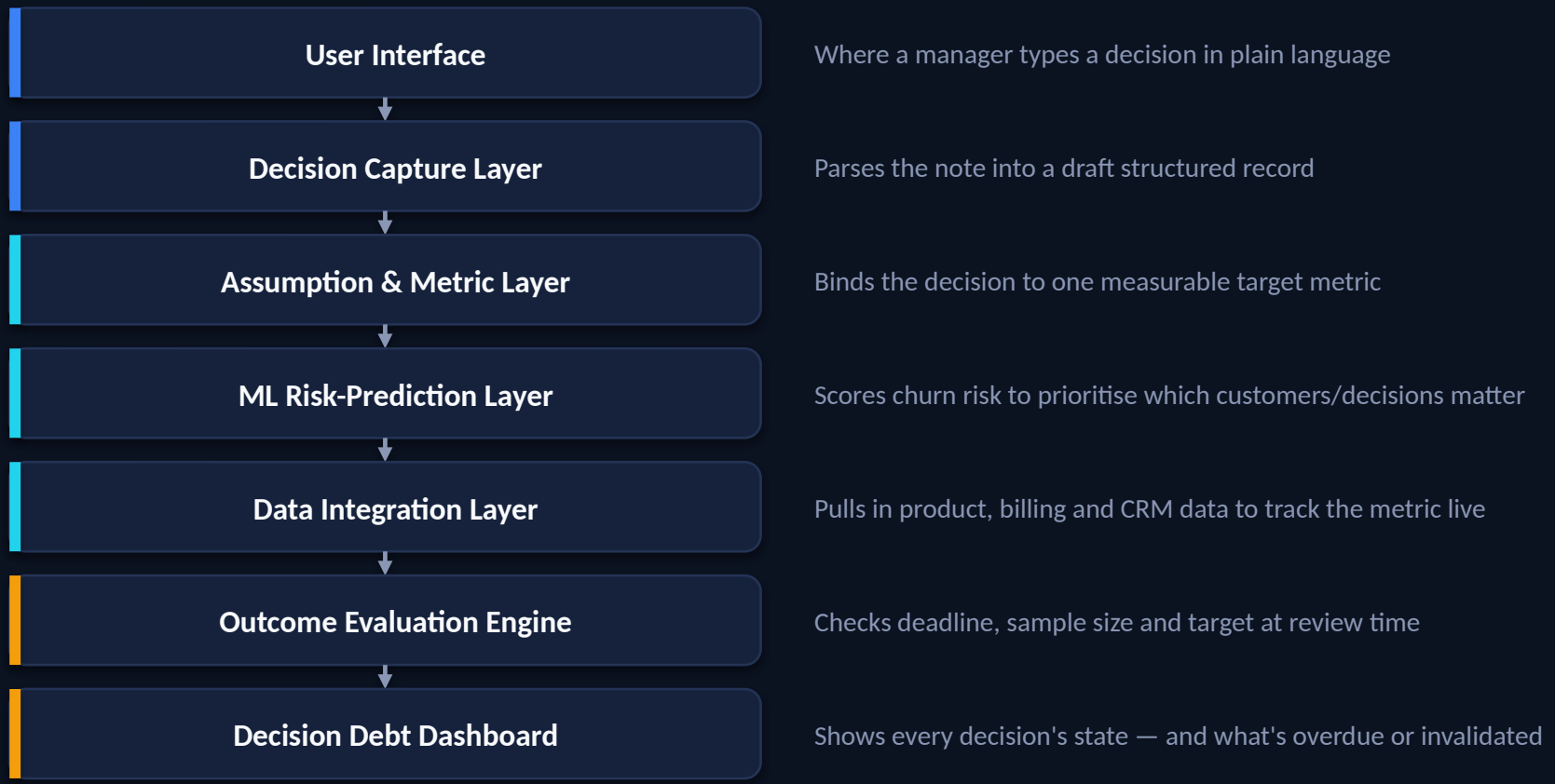
Top churn-risk drivers (feature importance)



A model that flags churn risk does not prove an intervention caused a customer to stay — prediction identifies candidates for a decision; it does not validate the decision's outcome.

SYSTEM ARCHITECTURE

Seven Layers, One Feedback Loop



BUILT WITH

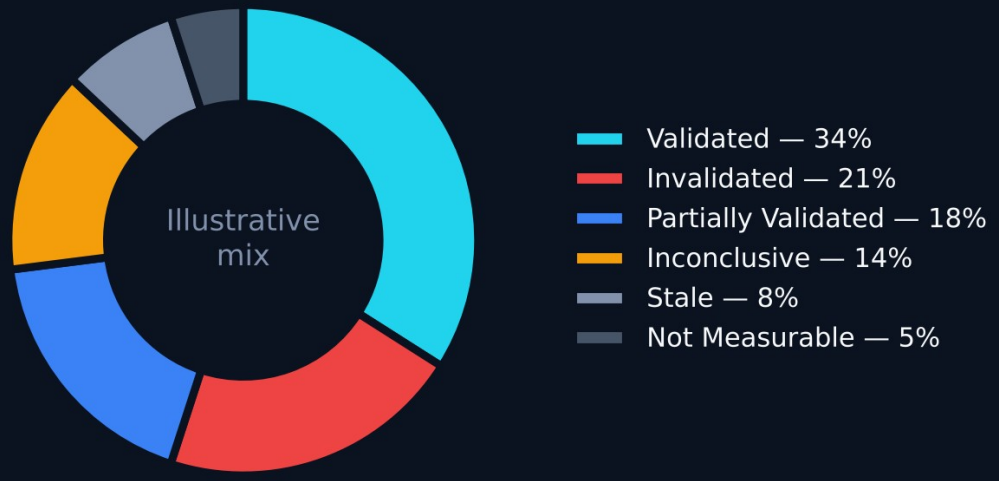
Python | Pandas | Scikit-learn | Streamlit | Plotly

DECISION EVALUATION ENGINE

Four Checks, Six Possible States

- Was the target metric reached?
- Has the review deadline passed?
- Was the sample size large enough to trust the result?
- Is the result statistically conclusive, or still noisy?

Illustrative state mix — simulated for demonstration



Validated / Invalidated = target clearly met or missed, with enough data. Partially Validated = directionally right, underpowered. Inconclusive = evidence too weak either way. Stale = deadline passed, never reviewed. Not Measurable = metric or data unavailable.

DIFFERENTIATION & PRIOR ART

Honest Positioning: The Pieces Exist. The Workflow Is New.

	Captures decisions	Explicit assumption	Forecasts outcome	Closes loop w/ measured outcome	Built for product & retention teams
BI dashboards (Looker, Tableau, Power BI)	–	–	–	–	–
Jira / Linear / Notion	◐	–	–	–	◐
Corporate prediction markets (P0, Hypermind)	–	◐	✓	◐	–
Investment decision trackers	✓	◐	◐	◐	–
Enterprise decision intelligence (Vigil, org-memory patents)	✓	✓	◐	◐	–
Decision Debt OS (this project)	✓	✓	✓	✓	✓

Differentiation = the lightweight decision-to-outcome workflow for product and retention teams — **not a new machine-learning technique, and not a claim that every component here is novel on its own.**

TARGET CUSTOMER & BUSINESS MODEL

B2B SaaS, 20–150 Employees, Active Growth Teams

BUYER

VP Product or Head of Growth / Retention at a venture-backed B2B SaaS company

INITIAL USE CASE

Churn-intervention decisions — discounts, win-back campaigns, onboarding changes

PILOT PRICE

$499 / month per workspace

EXPANSION PATH

Retention decisions → pricing & packaging → feature launches → org-wide decision registry

BUSINESS MODEL

B2B SaaS subscription, per-workspace, usage-based tiers above the pilot

Business hypothesis — not yet validated with paying customers or signed pilots.

RISKS & LIMITATIONS

What Could Go Wrong — And the Mitigation

RISK	MITIGATION
Synthetic data cannot prove real-world value	Pilot with design-partner companies' real (anonymised) data before any efficacy claim.
No guaranteed causal inference from churn prediction	Treat scores as risk triage only; require a held-out comparison group before crediting any intervention.
Metric-binding errors — wrong metric tied to a decision	Shared metric ontology plus reviewer sign-off before a decision record goes live.
Poor data quality / integration gaps	Data-quality checks at ingestion; low-confidence source feeds are flagged, not trusted silently.
Forecast manipulation — gaming confidence scores	Track forecaster calibration over time so gaming becomes visible, not just the raw score.
Privacy — customer PII inside decision records	Field-level encryption, role-based access, minimal PII in ML features.
False alerts / alert fatigue from risk scoring	Tunable thresholds and a batched digest instead of per-customer pings.
User resistance / adoption friction	Natural-language capture to avoid new process overhead; start with one team, not the whole org.
Prior-art & patent overlap (e.g., EP1402435A4, US20250217680A1)	Freedom-to-operate review before commercial launch; keep claims to the workflow, not the method.